Inventory optimization

A manufacturing company produces three types of products (Product A, Product B, and Product C). The production process occurs over four months (January, February, March, and April). Each product requires a certain amount of production time, incurs a certain production cost, and meets a specific monthly demand. Additionally, there is a limited number of production hours available each month. The company also incurs inventory holding costs for storing products from one month to the next. The objective is to minimize the total production and inventory holding costs while meeting the demand for each product.

Product	**Hours per unit**
Product A	1.2
Product B	1.3
Product C	1

Table - Production hours required per unit

Product	**Cost per unit (£)**
Product A	20
Product B	30
Product C	10

Table- Production cost per unit

Month	**Hours available**
January	500
February	450
March	400
April	350

Table - Monthly production hours available

Month	Product A	Product B	Product C
January	100	80	120
February	110	90	130
March	120	100	140
April	130	110	150

Table - Monthly demand for each product

Product	Holding cost (£)
Product A	1
Product B	2
Product C	0.5

Table - Inventory holding cost per unit per month (£)

Additional Constraints:

- Production in each month must meet or exceed the demand for that month.
- Inventory can be carried over from one month to the next to meet future demands.
- The company starts with zero inventory for all products in January.
- The total production hours used each month cannot exceed the available production hours.

(a) **Formulate an integer programming model and solve it to help the company optimise production and inventory planning.**

Problem statement:

Manufacturing company produces three product types: A, B and C over four months January to April. The company's aim is to reduce the overall production and inventory costs while meeting the monthly demand for each product. The problem is solved using Linear Programming (LP) with integer constraints, specifically Mixed-Integer Linear Programming (MILP) due to the integer variables for production and inventory.

Decision variables

P_{ijt}: number of units of product j for $j\{A, B, C\}$ produced in month t $t = \{1, 2, 3, 4\}$

I_{ijt}: Inventory of product j at the end of month $t = \{1, 2, 3, 4\}$

Objective function

Minimize the production cost & Inventory holding cost

$$Z = \sum_{j \in \{A, B, C\}} \sum_{t=1}^{4} (c_j \cdot P_{jt} + h_j \cdot I_{jt})$$

c_j = production cost per unit of product j

h_j = Inventory holding cost per unit of product j

Constraints

1.Sum of production and inventory carried each month should meet or exceed demand

Production hour constraint

$$P_{jt} + I_{j(t-1)} \geq D_{jt} + I_{jt}$$

for all $j = \{A, B, C\}$; $t = \{1, 2, 3, 4\}$

where,

D_{jt} = Demand for product j in month t.

$I_{j0} = 0$ Initial inventory in January is 0.

Total mathematical representation

$$\boldsymbol{Z = \sum_{t=1}^{4} [20 \cdot P_{At} + 30 \cdot P_{Bt} + 10 \cdot P_{Ct} + 1 \cdot I_{At} + 2 \cdot I_{Bt} + 0.5 \cdot I_{Ct}]}$$

Demand Constraints:

Production Capacity Constraints:

- **Month 1:**
  $$1.2P_{A1} + 1.3P_{B1} + P_{C1} \leq 500$$
- **Month 2:**
  $$1.2P_{A2} + 1.3P_{B2} + P_{C2} \leq 450$$
- **Month 3:**
  $$1.2P_{A3} + 1.3P_{B3} + P_{C3} \leq 400$$
- **Month 4:**
  $$1.2P_{A4} + 1.3P_{B4} + P_{C4} \leq 350$$

Initial Inventory:

$$I_{A0} = I_{B0} = I_{C0} = 0$$

Non negativity constraints:

$$P_{jt} \geq 0, \quad I_{jt} \geq 0, \quad \forall j, t$$

- **Product A:**
  $$P_{A1} = 100 + I_{A1}$$
  $$P_{A2} + I_{A1} = 110 + I_{A2}$$
  $$P_{A3} + I_{A2} = 120 + I_{A3}$$
  $$P_{A4} + I_{A3} = 130 + I_{A4}$$
- **Product B:**
  $$P_{B1} = 80 + I_{B1}$$
  $$P_{B2} + I_{B1} = 90 + I_{B2}$$
  $$P_{B3} + I_{B2} = 100 + I_{B3}$$
  $$P_{B4} + I_{B3} = 110 + I_{B4}$$
- **Product C:**
  $$P_{C1} = 120 + I_{C1}$$
  $$P_{C2} + I_{C1} = 130 + I_{C2}$$
  $$P_{C3} + IC2 = 140 + I_{C3}$$
  $$P_{C4} + I_{C3} = 150 + I_{C4}$$

Solution:

Production Plan			
Month	Product A	Product B	Product C
January	100	80	162
February	110	90	201
March	120	100	126
April	130	110	51

Inventory levels			
Month	Product A	Product B	Product C
January	0	0	42
February	0	0	113
March	0	0	99
April	0	0	0

The optimal value is: £26,127.

(b) Assume that the production cost for Product B increases by £5 per unit in March due to increased raw material costs. Rebuild the model and solve it under the new conditions.

Solution:

Production Plan			
Month	Product A	Product B	Product C
January	100	80	179
February	110	190	71
March	120	0	239
April	130	110	51

Inventory levels			
Month	Product A	Product B	Product C
January	0	0	59
February	0	100	0
March	0	0	99
April	0	0	0

The optimal value is: £26,279.

(c) Due to a sudden increase in demand, the demand for Product C in April increases by 20 units. Rebuild the model and solve it.

Solution:

Production Plan			
Month	Product A	Product B	Product C
January	100	80	182
February	110	90	201
March	120	100	126
April	130	110	51

Inventory levels			
Month	Product A	Product B	Product C
January	0	0	62
February	0	0	133
March	0	0	119
April	0	0	0

The optimal value is: £26,357.

(d) The company wants to ensure that the total inventory holding cost does not exceed £500 over the entire planning period. Rebuild the model and solve it.

Solution:

Production Plan			
Month	**Product A**	**Product B**	**Product C**
January	100	80	162
February	110	90	201
March	120	100	126
April	130	110	51

Inventory levels			
Month	**Product A**	**Product B**	**Product C**
January	0	0	42
February	0	0	113
March	0	0	99
April	0	0	0

The optimal value is: £26,127.

Insights and Recommendations:

(a) The production and inventory planning model are resilient to minor changes:

Increase in Product B's Cost: A £5 increase per unit in March leads to a minimal cost increase of £150, indicating that the model is not highly sensitive to Product B's cost changes.

Increase in Demand for Product C: The total cost has risen by £78, which indicates that the £20 hike in the price of each unit in April can increase only this much total cost. This suggests that demand variations can be met without the organization incurring heavy expenses.

Holding Cost Limit of £500: The company manages the inventory well, and using the inventory carrying cost constraint has no impact on the total cost, which shows effective inventory management.

Generally, the company's optimization model is adaptable and highly efficient, which can tolerate some small variations of cost and demand and even keep the company profitable. The company is already managing its inventory costs well and can handle market shifts with minor adjustments.